

Chuyue Li

(+1)217-200-3589 | lichuyue0312@gmail.com (preferred) | chuyuel2@illinois.edu

Research Interest: Agentic RL, Multimodal Generation/Understanding/Reasoning, Self-Evolve Agents

Education

University of Illinois Urbana-Champaign MS of Computer Science(MSCS)	2026.08 - 2028.06
University of California, Berkeley CS Junior <i>Exchange</i> GPA: 3.9/4.0	2024.08 - 2025.05
ShanghaiTech University CS <i>Undergrad</i> Major GPA 3.87/4.0	2022.9 - 2026.7

Internship Experience

Alibaba Qwen | Research Intern 2026.02-2026.08

Self-Evolving Agent for Video Understanding and Representation Learning

Visual Technology Department, AIGC Team Mentor: Jingpeng Yu, Jiaming Liu

- **First-Authored Paper Under Review:** AVA-Encoder: Towards Agent-Native Video Representation Learning.
- Served as first author; pioneered agentic video auto-encoding by representing videos as a structured Knowledge Graph. Enabled agents to encode videos into structured graphs and reconstruct them via a fixed decoder, optimizing the system through textual-gradient feedback based on reconstruction consistency.
- Designed a Hierarchical Agentic Encoder to minimize information loss. Developed a dual-loop optimization mechanism: an Outer Loop with an Anti-Forgetting Gate to learn transferable encoding policies, and an Inner Loop with an Anti-Degradation Gate for test-time KG refinement.
- Achieved SOTA reconstruction fidelity (49.0% overall, a 20.7% absolute gain over baseline). Notably, the self-learned policy outperformed human-tuned version (45.8% vs. 44.4%) while reducing token consumption by 74.3% (shot level) and 70.1% (keyframe level), enabling highly efficient, topology-aware editing.

Tencent Hunyuan | Research Intern 2025.06-2025.11

Multi-level Auto-Planning Interactive Visual Generative Agent

Multimodal Model Department, Visual Agent Team Mentor: Rongwei Quan

- **Accepted by CVPR 2026 Findings:** VisionCreator: A Native Visual-Generation Agentic Model with Understanding, Thinking, Planning and Creation.
- Built a long-horizon visual generation agent that autonomously decomposes complex image/video requests, plans multi-step workflows, invokes generation/editing tools, and self-adjusts execution across 20+ step trajectories.
- Developed the agentic post-training pipeline with Progressive Specialization Training and GRPO-based Virtual RL; constructed a 36-tool simulated environment to replace costly real image/video generation during RL, improving the initialization reward by +0.23 and reducing RL convergence time by 50%.
- Designed plan-level + execution-level trajectory rewards to optimize tool calling, structural correctness, result achievement, and cross-shot visual consistency; achieved +49% over the strongest SFT/PST baseline and outperformed GPT-5 and Gemini-2.5-Pro in human evaluation (3.42 vs. 3.19 / 3.01).

Circle Cat Research Department | Research Intern USA 2025.2 - China 2026.02

LLM for Multi-level Coding Error Classification

Role: Project Leader, First Author Supervisor: Kazuma Hashimoto, Lingyu Gao

- **Accepted by NeurIPS 2026 TAE:** PyMETA: A Benchmark Dataset for Hierarchical Student Code Error Classification with Python-Interpreter-Based Labels
- Design a multi-level, fine-grained taxonomy for web-environment Python code error types in online compilers with LLM integration, and develop a benchmark based on 40,000 real-world Python code errors with multi-level classification, combining single-error and multi-error detection cases.
- Enhance LLM's multi-level error detection and classification through fine-tuning and prompting on multiple models, leveraging real error patterns and unstructured debugging logs from online compilers.
- Analyze and compare LLM performance on hierarchical error granularity and single-to-multi error detection tasks, examining biases, capability limits, and confidence across seven model architectures.

Multi-modal Time Series Conditional Generation from Unstructured Text***Supervisor: Kan Ren***

- **Accepted by ICML2025:** VerbalTS: Generating Time Series from Texts
- Designed a **multi-focal framework** for finer-grained cross-modal semantic alignment and controllable time-series generation from unstructured text; validated on **6 synthetic and real-world datasets**.
- Implemented a **Multi-view Noise Estimator** that performs multi-resolution modeling and noise estimation across **temporal, spatial, and diffusion** perspectives, achieving **20%+ fidelity improvements** on multiple datasets.
- Proposed a **Multi-focal Text Processor** that decomposes text embeddings into multi-semantic levels across temporal resolutions, spatial variables, and diffusion stages, using **learnable anchor vectors and semantic reprogramming** for fine-grained text alignment.
- Built an **adapter module** that progressively integrates and aligns multi-semantic textual information with the time-series generation process for precise, hierarchical cross-modal control, reducing Weather J-FTSD from **18.33 to 8.56**.
- Proposed the **CTTP model and CTTP Score** for text-time-series semantic evaluation; improved **FID, J-FTSD, and CTTP** by **20%+** over strong baselines, with up to **58.7% lower FID**.

Unified Model for Multi-modal TS Generation, Prediction and Editing***Supervisor: Kan Ren***

- **Under review:** Unified Generative Modeling for Multimodal Time Series Analysis
- Propose GENTS: unifying time series conditional generation, forecasting, and editing into a masked-conditional generation problem, modeled within a unified framework.
- Design the unified noise estimator enabling cross-task generative capabilities and adaptive multimodal time series modeling.
- Introduce a domain-agnostic automated time series captioning method based on the statistical and morphological features of time series data, effectively addressing the scarcity of domain-specific multimodal paired data.

Publications & Manuscripts

-
- [1] AVA-Encoder: Towards Agent-Native Video Representation Learning. **Under Review**
[Chuyue Li, Jinpeng Yu, Haozhe Wang, Tian Xueyun, Zhijing Zhang, Bingnan Li, Shuqi Gu, Kan Ren, Jiaming Liu, Ruihua Huang]
- [2] VisionCreator: A Native Visual-Generation Agentic Model with Understanding, Thinking, Planning and Creation. **Accepted by CVPR2026 Findings**.
[Jinxiang Lai, Zexin Lu, Jiajun He, Rongwei Quan, Wenzhe Zhao, Qinyu Yang, qi chen, Qin Lin, **Chuyue Li**, Tao Gao, Song Guo, Qinglin Lu]
- [3] PyMETA: A Benchmark Dataset for Hierarchical Student Code Error Classification with Python-Interpreter-Based Labels. **Accepted by NeurIPS 2026 TAE**
[**Chuyue Li**, Ziqi Tang, Jingyi Wang, Yu Wu, Kazuma Hashimoto, Lingyu Gao]
- [4] What to Edit Next: Visually Aligned Image-Editing Follow-Up Suggestions in Conversational Systems
[Zhijing Zhang, Jinpeng Yu, Xin Song, Bingnan Li, **Chuyue Li**, Changhui Du, Xiaolin Fang, Jiaming Liu, Ruihua Huang]
- [5] VerbalTS: Generating Time Series from Texts. **Accepted by ICML2025**
[Shuqi Gu, **Chuyue Li**, Baoyu Jing, Kan Ren]
- [6] EvoScript: Reinforcement Learning for Script Generation with Evolving Rubrics. **Under review**.
[Tian Xueyun, Bingbing Xu **Chuyue Li**, Jiaming Liu, Chao Zhang, Weijia Feng, MingHua Ma, Yu Zhao, Jingchu Wang, Ruihua Huang, Yuanzhuo Wang, Huawei Shen]
- [7] Learning to Simulate Individuals from Macro Social Signals. **Under review**.
[Yining Zhao, Bushi Liu, Haofei Yu, Zhengyang Qi, Shanyong Wang **Chuyue Li** Yuxiang Liu, Jiaxuan You]
- [8] Unified Generative Modeling for Multimodal Time Series Analysis. **Under review**.
[Shuqi Gu, **Chuyue Li**, Yi Huang, Baoyu Jing, Yongxiang Zhao, Kan Ren]